

Mini_Project5

October 3, 2026

```
[16]: pip install pandas
```

```
Requirement already satisfied: pandas in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2026.4)
Requirement already satisfied: tzdata>=2022.7 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2026.4)
Requirement already satisfied: six>=1.5 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[17]: import pandas as pd
```

```
[18]: data=pd.read_csv(r"C:\Users\surface\Downloads\house_prices_practice.csv")
```

```
[19]: data.head()
```

```
[19]:
```

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	\
0	1	7	1560	0	1658	1969	2	
1	2	4	2827	2	1319	2012	3	
2	3	8	3920	0	841	2010	1	
3	4	5	3044	0	1058	1998	0	
4	5	7	801	1	2428	2020	0	

	BedroomAbvGr	LotArea	SalePrice
0	1	8059	177106
1	4	13530	301044
2	4	9010	360609

3	4	13207	240556
4	1	9117	193656

```
[20]: data.tail()
```

```
[20]:
```

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	\
295	296	1	3495	1	1792	1954	
296	297	5	3438	3	1266	2003	
297	298	6	1992	0	1148	1996	
298	299	3	3722	1	1407	1998	
299	300	9	1667	0	988	1958	

	FullBath	BedroomAbvGr	LotArea	SalePrice
295	2	5	4978	250604
296	0	1	9373	329906
297	1	1	7907	184623
298	1	1	8097	303345
299	2	1	14553	204982

```
[21]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id              300 non-null   int64
1   OverallQual     300 non-null   int64
2   GrLivArea       300 non-null   int64
3   GarageCars      300 non-null   int64
4   TotalBsmtSF     300 non-null   int64
5   YearBuilt       300 non-null   int64
6   FullBath        300 non-null   int64
7   BedroomAbvGr    300 non-null   int64
8   LotArea         300 non-null   int64
9   SalePrice       300 non-null   int64
dtypes: int64(10)
memory usage: 23.6 KB
```

```
[22]: data.describe()
```

```
[22]:
```

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	\
count	300.000000	300.000000	300.000000	300.000000	300.000000	
mean	150.500000	5.326667	2307.386667	1.330000	1468.796667	
std	86.746758	2.873001	1042.561303	1.109898	672.333705	
min	1.000000	1.000000	504.000000	0.000000	303.000000	
25%	75.750000	3.000000	1392.250000	0.000000	903.000000	
50%	150.500000	5.000000	2265.500000	1.000000	1502.000000	

75%	225.250000	8.000000	3306.500000	2.000000	2129.500000
max	300.000000	10.000000	3998.000000	3.000000	2492.000000

	YearBuilt	FullBath	BedroomAbvGr	LotArea	SalePrice
count	300.000000	300.000000	300.000000	300.000000	300.000000
mean	1986.163333	1.523333	2.926667	8969.453333	252262.903333
std	21.377089	1.131543	1.456604	3753.531132	74998.055214
min	1950.000000	0.000000	1.000000	2009.000000	82494.000000
25%	1967.000000	0.000000	2.000000	5996.250000	190355.250000
50%	1986.000000	2.000000	3.000000	9031.000000	251292.500000
75%	2004.250000	3.000000	4.000000	12316.000000	307105.000000
max	2023.000000	3.000000	5.000000	14987.000000	435291.000000

```
[23]: data.shape
```

```
[23]: (300, 10)
```

```
[24]: data.columns
```

```
[24]: Index(['Id', 'OverallQual', 'GrLivArea', 'GarageCars', 'TotalBsmtSF',
        'YearBuilt', 'FullBath', 'BedroomAbvGr', 'LotArea', 'SalePrice'],
        dtype='object')
```

```
[25]: data.dtypes
```

```
[25]: Id                int64
OverallQual          int64
GrLivArea            int64
GarageCars           int64
TotalBsmtSF          int64
YearBuilt            int64
FullBath             int64
BedroomAbvGr         int64
LotArea              int64
SalePrice            int64
dtype: object
```

```
[26]: data.isnull().sum()
```

```
[26]: Id                0
OverallQual          0
GrLivArea            0
GarageCars           0
TotalBsmtSF          0
YearBuilt            0
FullBath             0
BedroomAbvGr         0
LotArea              0
```

```
SalePrice      0
dtype: int64
```

```
[27]: int(data.duplicated().sum())
```

```
[27]: 0
```

```
[28]: data.drop_duplicates(inplace=True)
```

```
[29]: data.nunique()
```

```
[29]: Id                300
OverallQual          10
GrLivArea           287
GarageCars           4
TotalBsmtSF         278
YearBuilt            73
FullBath             4
BedroomAbvGr         5
LotArea             296
SalePrice            300
dtype: int64
```

```
[32]: x=data[['OverallQual','GrLivArea','GarageCars','TotalBsmtSF',
            'YearBuilt','FullBath','BedroomAbvGr','LotArea']]
y=data[['SalePrice']]
print(x)
print(y)
```

	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	\
0	7	1560	0	1658	1969	2	
1	4	2827	2	1319	2012	3	
2	8	3920	0	841	2010	1	
3	5	3044	0	1058	1998	0	
4	7	801	1	2428	2020	0	
..	...	...	...	...	...	...	
295	1	3495	1	1792	1954	2	
296	5	3438	3	1266	2003	0	
297	6	1992	0	1148	1996	1	
298	3	3722	1	1407	1998	1	
299	9	1667	0	988	1958	2	

	BedroomAbvGr	LotArea
0	1	8059
1	4	13530
2	4	9010
3	4	13207
4	1	9117

```

..      ...
295      5      4978
296      1      9373
297      1      7907
298      1      8097
299      1     14553

```

[300 rows x 8 columns]

```

SalePrice
0      177106
1      301044
2      360609
3      240556
4      193656
..      ...
295     250604
296     329906
297     184623
298     303345
299     204982

```

[300 rows x 1 columns]

```

[33]: from sklearn.model_selection import train_test_split
      x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,
      ↪random_state=12)

```

```

[34]: from sklearn.linear_model import LinearRegression
      slr=LinearRegression()
      slr.fit(x_train,y_train)
      slr.predict(x_test)

```

```

[34]: array([[322282.5431834 ],
             [385458.15265493],
             [207237.33477461],
             [330855.22338302],
             [121908.96253398],
             [375108.10558674],
             [314546.53143914],
             [208021.40165097],
             [195653.48014182],
             [263678.98794911],
             [301812.54037438],
             [193196.30280169],
             [288125.8293911 ],
             [287327.13221277],
             [279901.60413633],

```

[259500.11596118],
[244544.14681399],
[92726.42536953],
[195930.78396675],
[380829.69796294],
[186627.69311551],
[226935.83169137],
[236240.71994047],
[163129.99885131],
[253733.72020922],
[251503.62688526],
[209578.9074317],
[330829.30525549],
[174989.22697722],
[165515.57350818],
[169754.92556337],
[219836.21560033],
[242926.15981885],
[219144.61969513],
[395918.92166136],
[252044.24341115],
[280225.93086959],
[378973.90634702],
[148145.86406382],
[317629.90704935],
[302248.44565522],
[143897.85285476],
[169141.58142888],
[315231.93304144],
[216759.68204755],
[297999.35376512],
[257229.93545587],
[244289.26222301],
[340693.30193849],
[368655.09251521],
[194289.10836118],
[269220.63257097],
[192068.90251095],
[210628.44545573],
[292775.83192997],
[115431.99476816],
[206191.59940719],
[282692.11991387],
[118876.94713104],
[169476.9318799]])

```
[36]: x=data[['OverallQual','GrLivArea','GarageCars','TotalBsmtSF',
            'YearBuilt','FullBath','BedroomAbvGr','LotArea']]
y=data[['SalePrice']]
print(x)
print(y)
```

	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	\
0	7	1560	0	1658	1969	2	
1	4	2827	2	1319	2012	3	
2	8	3920	0	841	2010	1	
3	5	3044	0	1058	1998	0	
4	7	801	1	2428	2020	0	
..	...	...	...	...	...	...	
295	1	3495	1	1792	1954	2	
296	5	3438	3	1266	2003	0	
297	6	1992	0	1148	1996	1	
298	3	3722	1	1407	1998	1	
299	9	1667	0	988	1958	2	

	BedroomAbvGr	LotArea
0	1	8059
1	4	13530
2	4	9010
3	4	13207
4	1	9117
..	...	...
295	5	4978
296	1	9373
297	1	7907
298	1	8097
299	1	14553

[300 rows x 8 columns]

	SalePrice
0	177106
1	301044
2	360609
3	240556
4	193656
..	...
295	250604
296	329906
297	184623
298	303345
299	204982

[300 rows x 1 columns]

```
[37]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,
↳random_state=12)
```

```
[38]: from sklearn.linear_model import LinearRegression
slr=LinearRegression()
slr.fit(x_train,y_train)#Model Training
slr.predict(x_test) #model testing
```

```
[38]: array([[322282.5431834 ],
[385458.15265493],
[207237.33477461],
[330855.22338302],
[121908.96253398],
[375108.10558674],
[314546.53143914],
[208021.40165097],
[195653.48014182],
[263678.98794911],
[301812.54037438],
[193196.30280169],
[288125.8293911 ],
[287327.13221277],
[279901.60413633],
[259500.11596118],
[244544.14681399],
[ 92726.42536953],
[195930.78396675],
[380829.69796294],
[186627.69311551],
[226935.83169137],
[236240.71994047],
[163129.99885131],
[253733.72020922],
[251503.62688526],
[209578.9074317 ],
[330829.30525549],
[174989.22697722],
[165515.57350818],
[169754.92556337],
[219836.21560033],
[242926.15981885],
[219144.61969513],
[395918.92166136],
[252044.24341115],
[280225.93086959],
[378973.90634702],
```

```
[148145.86406382],  
[317629.90704935],  
[302248.44565522],  
[143897.85285476],  
[169141.58142888],  
[315231.93304144],  
[216759.68204755],  
[297999.35376512],  
[257229.93545587],  
[244289.26222301],  
[340693.30193849],  
[368655.09251521],  
[194289.10836118],  
[269220.63257097],  
[192068.90251095],  
[210628.44545573],  
[292775.83192997],  
[115431.99476816],  
[206191.59940719],  
[282692.11991387],  
[118876.94713104],  
[169476.9318799 ]])
```

```
[39]: from sklearn.metrics import r2_score  
y_pred=slr.predict(x_test)  
r2=r2_score(y_test,y_pred)  
print(r2)
```

```
0.9776064358983928
```